

Linux Fundamentals

Navigation Commands

pwd - Shows current directory

ls - Lists files and folders

cd - Change directory

mkdir test - Create a new directory

touch file.txt - Create a new file

rm file.txt - Delete a file

User Commands

whoami - Shows current user

Networking Commands

ifconfig - Display network configuration

ip a - Show IP address information

ping google.com - Test connectivity

netstat - Display network connections

traceroute google.com - Show route to destination

Package Management

sudo apt update - Update package list

sudo apt upgrade - Upgrade installed packages

dpkg -l - List installed packages

Permissions

chmod 755 file.txt - Change permissions

chown user:user file.txt - Change ownership

```
(kali@kali)-[~]
└─$ pwd
/home/kali

(kali@kali)-[~]
└─$ ls
Desktop  Downloads  Pictures  Templates  Videos
Documents Music      Public    test

(kali@kali)-[~]
└─$ ls -l
Desktop
Documents
Downloads
Music
Pictures
Public
Templates
test
Videos

(kali@kali)-[~]
└─$ ls -la
total 168
drwxr-xr-x 17 kali kali 4096 Jun  2 05:46 .
drwxr-xr-x  3 root root 4096 Mar 20 02:45 ..
-rw-r--r--  1 kali kali  220 Mar 20 02:45 .bash_logout
-rw-r--r--  1 kali kali 5578 Mar 20 02:45 .bashrc
-rw-r--r--  1 kali kali 3526 Mar 20 02:45 .bashrc.original
drwxr-xr-x  8 kali kali 4096 Jun  2 05:34 .cache
drwxr-xr-x 13 kali kali 4096 Jun  2 05:34 .config
drwxr-xr-x  2 kali kali 4096 Jun  2 05:32 Desktop
-rw-r--r--  1 kali kali  35 Jun  2 05:32 .dmrc
drwxr-xr-x  2 kali kali 4096 Jun  2 05:32 Documents
drwxr-xr-x  2 kali kali 4096 Jun  2 05:32 Downloads
-rw-r--r--  1 kali kali 11759 Mar 20 02:45 .face
lrwxrwxrwx  1 kali kali  5 Mar 20 02:45 .face.icon -> .face
drwxr-xr-x  3 kali kali 4096 Jun  2 05:32 .gnupg
-rw-r--r--  1 kali kali  0 Jun  2 05:32 .ICEauthority
drwxr-xr-x  3 kali kali 4096 Mar 20 02:45 .java
drwxr-xr-x  5 kali kali 4096 Jun  2 05:32 .local
drwxr-xr-x  2 kali kali 4096 Jun  2 05:32 Music
drwxr-xr-x  2 kali kali 4096 Jun  2 05:44 Pictures
-rw-r--r--  1 kali kali  807 Mar 20 02:45 .profile
drwxr-xr-x  2 kali kali 4096 Jun  2 05:32 Public
drwxr-xr-x  3 kali kali 4096 Jun  2 05:32 .ssh
drwxr-xr-x  2 kali kali 4096 Jun  2 05:32 Templates
```

```
(kali@kali)-[~/Desktop]
└─$ mkdir CyberLab

(kali@kali)-[~/Desktop]
└─$ cd CyberLab

(kali@kali)-[~/Desktop/CyberLab]
└─$ touch notes.txt

(kali@kali)-[~/Desktop/CyberLab]
└─$ ls -l
total 0
-rw-rw-r-- 1 kali kali 0 Jun  2 06:01 notes.txt

(kali@kali)-[~/Desktop/CyberLab]
└─$ chmod 777 notes.txt

(kali@kali)-[~/Desktop/CyberLab]
└─$ chmod 600 notes.txt

(kali@kali)-[~/Desktop/CyberLab]
└─$ whoami
kali

(kali@kali)-[~/Desktop/CyberLab]
└─$ sudo chown kali:kali notes.txt
[sudo] password for kali:

(kali@kali)-[~/Desktop/CyberLab]
└─$ ls -l
notes.txt

(kali@kali)-[~/Desktop/CyberLab]
└─$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.1 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [106 kB]
```

```
Session Actions Edit View Help
kali@kali: ~/Desktop/CyberLab
$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1509 packages can be upgraded. Run 'apt list --upgradable' to see them.

kali@kali:~/Desktop/CyberLab
$ apt search nmap
nmap/kali-rolling 0.20150903-r2013-10-b2 amd64
  genome assembly-to-assembly comparison

brutenspray/kali-rolling 2.2.2-0kali1 amd64
  Bruteforcing from various scanner output

nscan/kali-rolling 0.1.5-0kali1 all
  wrapper around nmap

nscan-doc/kali-rolling 0.1.5-0kali1 all
  wrapper around nmap (documentation)

nvmutils/kali-rolling 20230925.1-git20230720-0kali4 amd64
  Rapid web application triage tool

fierce/kali-rolling,now 1.6.0-2 all [installed,automatic]
  Domain DNS scanner

golang-github-lair-framework-go-nmap-dev/kali-rolling 0.0-git20191202.350700
  nmap XML parsing library for Go (library)

golang-github-malfunkt-iprange-dev/kali-rolling 0.9.0-5 all
  IPv4 address parser for the nmap format

incus-extra/kali-rolling 7.0.0-1 amd64
  Modern container and virtual machine manager - extra tools

ivre/kali-rolling 0.9.21-0kali1 all
  network recon framework IVRE of DRUNK

ivre-doc/kali-rolling 0.9.21-0kali1 all
  Documentation for IVRE package

ipfile-map-perl/kali-rolling 0.71-2 amd64
  Perl module providing simple and safe memory mapping

libnmap-parser-perl/kali-rolling 1.37-2 all
  module to parse nmap scan results with perl

librust-cpp-synmap-dev/kali-rolling 0.3.0-1+b3 amd64
  Sourcemap and full crate parsing support for 'cpp_syn' - Rust source code

lm-tools-nmap/kali-rolling 0.0-git20250612.36818ca-0kali1 all
  plugin for LHM tool hat provides Nmap network scanning capabilities
```

```
Session Actions Edit View Help
kali@kali:~/Desktop/CyberLab
Purging configuration files for lepton (0.5.2-0kali1)~.

kali@kali:~/Desktop/CyberLab
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::d83d:853e:feaa:806e prefixlen 64 scopeid 0x0<link>
    inet6 fd17:625c:f037:2:d579:b72e:b425:5642 prefixlen 64 scopeid 0x0
    <global> ether 08:00:27:8a:35:d2 txqueuelen 1000 (Ethernet)
    RX packets 55944 bytes 78603881 (74.9 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9354 bytes 589297 (575.4 KiB)
    TX errors 0 dropped 11 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 400 (400.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

kali@kali:~/Desktop/CyberLab
$ netstat -tln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State

kali@kali:~/Desktop/CyberLab
$ netstat -rn
kernel IP routing table
Destination      Gateway            Genmask           Flags  MSS  Window  irtt  Ifac
0 0.0.0.0         10.0.2.2          0.0.0.0           UG    0 0      0     eth0
10.0.2.0         0.0.0.0           255.255.255.0     U     0 0      0     eth0

kali@kali:~/Desktop/CyberLab
$ traceroute google.com
traceroute to google.com (142.251.223.110), 30 hops max, 60 byte packets
 1 10.0.2.2 (10.0.2.2)  1.030 ms  0.643 ms  0.496 ms
 2 * * *
 3 * * *
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 * * *
 9 * * *
```